

LC-3 Instruction Encodings

Arith./Logic	ADD [†]	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
		0	0	0	1	DR		SR1		0			0	0	SR2						
	ADD [†]	0	0	0	1	DR		SR1		1	imm5										
	AND [†]	0	1	0	1	DR		SR1		0			0	0	SR2						
	AND [†]	0	1	0	1	DR		SR1		1	imm5										
	NOT [†]	1	0	0	1	DR		SR		1			1	1	1	1	1				
Load/Store	LD [†]	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
		0	0	1	0	DR		PCoffset9													
	ST	0	0	1	1	SR		PCoffset9													
	LDI [†]	1	0	1	0	DR		PCoffset9													
	STI	1	0	1	1	SR		PCoffset9													
	LDR [†]	0	1	1	0	DR		BaseR		offset6											
	STR	0	1	1	1	SR		BaseR		offset6											
Branch	JMP	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
		1	1	0	0	0		0	0	BaseR		0						0	0	0	
	BR	0	0	0	0	cn	cz	cp	PCoffset9												
	JSRR	0	1	0	0	0		0	0	BaseR		0						0	0	0	
	JSR	0	1	0	0	1	PCoffset11														
Misc.	LEA	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
		1	1	1	0	DR		PCoffset9													
	TRAP	1	1	1	1	0			0	0	0	trapvect8									
	RTI	1	0	0	0	0												0	0	0	0
	<i>illegal</i>	1	1	0	1	x			x	x	x	x	x	x	x	x	x				

[†] Updates condition code (CC)

LC-3 Datapath Control Signals

(Use these for filling out LC-3 FSM state transition diagrams)

Signals to assert:

- LD.MAR
- LD.MDR
- LD.IR
- LD.REG
- LD.CC
- GatePC
- GateMDR
- GateALU
- GateMARMUX
- MEM.EN

Signals with named values:

- PCMUX: PC+1, BUS, ADDER
- SR1MUX: IR[11:9], IR[8:6], 6
- DRMUX: IR[11:9], 7, 6
- ALUK: ADD, AND, NOTA, PASSA
- ADDR1MUX: PC, BaseR
- ADDR2MUX: ZERO, offset6, PCoffset9, PCoffset11
- MARMUX: IR[7:0], ADDER

LC-3 Microcode Control Signal Bit Assignments

(Use these for filling out bits in *microcode tables*)

- PCMUX:
 - $00_2 \rightarrow \text{PC}+1$
 - $01_2 \rightarrow \text{BUS}$
 - $10_2 \rightarrow \text{ADDER}$
- SR1MUX:
 - $00_2 \rightarrow \text{IR}[11:9]$
 - $01_2 \rightarrow \text{IR}[8:6]$
 - $10_2 \rightarrow 6$
- DRMUX:
 - $00_2 \rightarrow \text{IR}[11:9]$
 - $01_2 \rightarrow 7$
 - $10_2 \rightarrow 6$
- ALUK:
 - $00_2 \rightarrow \text{ADD}$
 - $01_2 \rightarrow \text{AND}$
 - $10_2 \rightarrow \text{NOTA}$
 - $11_2 \rightarrow \text{PASSA}$
- ADDR1MUX:
 - $0 \rightarrow \text{PC}$
 - $1 \rightarrow \text{BaseR}$
- ADDR2MUX:
 - $00_2 \rightarrow \text{ZERO}$
 - $01_2 \rightarrow \text{offset6}$
 - $10_2 \rightarrow \text{PCoffset9}$
 - $11_2 \rightarrow \text{PCoffset11}$
- MARMUX:
 - $0 \rightarrow \text{IR}[7:0]$
 - $1 \rightarrow \text{ADDER}$

